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ITA0609

Exp-5 : Write a program for Implementation of K-Nearest Neighbours (K-NN) in Python.

```
# Import required libraries

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

# Load the Iris dataset

iris = load_iris()

X = iris.data    # Features
y = iris.target  # Target labels

# Split the dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create the K-NN classifier (k = 3)
knn = KNeighborsClassifier(n_neighbors=3)

# Train the model
knn.fit(X_train, y_train)

# Predict the test data
y_pred = knn.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Display results
print("Predicted Labels:", y_pred)
print("Actual Labels  :", y_test)
print("Accuracy:", accuracy * 100, "%")
```

Output:

```
... Predicted Labels: [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
   Actual Labels   : [1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0]
   Accuracy: 100.0 %
```